Physics of Complex Systems

Projects

Course 2025-2026

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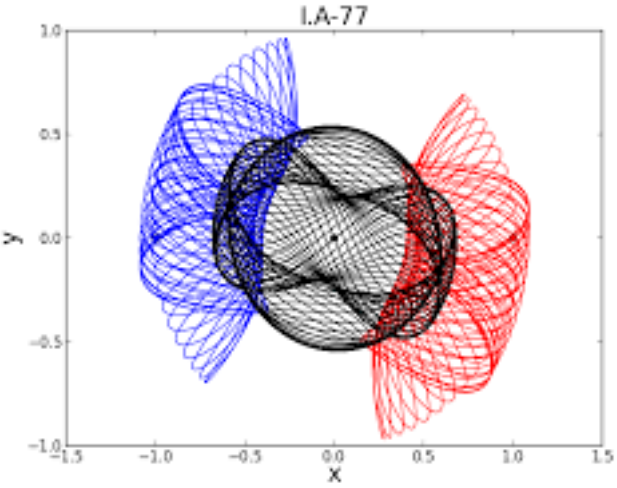
1 Chaos in classical mechanics. The Three-Body Problem

The motion of three point masses m_1, m_2, and m_3 interacting via Newtonian gravity in a 2D plane is governed by the following system of second-order differential equations:

$$\ddot{\mathbf{r}}_i = -G \sum_{j \neq i} \frac{m_j(\mathbf{r}_i - \mathbf{r}_j)}{\|\mathbf{r}_i - \mathbf{r}_j\|^3}, \quad i, j \in \{1, 2, 3\}$$

This system is described by a set of first-order Ordinary Differential Equations (ODEs) based on Newton's Equations. In this project, we will restrict ourselves to three point masses (m_1, m_2, m_3) in a 2D plane. In the Circular Restricted Three-Body Problem, meaning that orbits are circular and $m_3 \sim 0$ we can identify the five Lagrangian equilibrium points (L_1–L_5).



To study more general problems (but still in $2D$) we may implement a high-order integrator (e.g., RK4) and use energy conservation ($\Delta H/H_0$) over $T > 10^3$ orbital periods to ensure the validity of the results. The stability of initial conditions can be then calculated by the Lyapunov exponent.

$$\lambda = \lim_{t \to \infty} \frac{1}{t} \ln \frac{\|\Delta \mathbf{x}(t)\|}{\|\Delta \mathbf{x}(0)\|}$$

where $\Delta \mathbf{x}$ represents a small perturbation in the phase space vector. To identify the Lyapunov time scale. we may plot $\log \|\Delta \mathbf{x}\|$ vs. time.

1.1 Deliverables

- Identify the five Lagrangian equilibrium points (L_1–L_5). Perform a linear stability analysis of the L_4 and L_5 points. Determine the critical mass ratio ($\mu = \frac{m_1}{m_2}$) required for these points to remain stable.
- Find at least five closed trajectories of the three-body problem by parameter exploration.
- Calculate the Lyapunov exponents of initial conditions close to the fixed points and arbitrary ones.

2 Quantum Chaos and the Statistics of Energy Levels

In classical mechanics, chaos is defined by exponential sensitivity to initial conditions. In quantum mechanics, the linearity of the Schrödinger equation prevents this. Instead, "Quantum Chaos" manifests in the statistical properties of the energy spectrum. This project explores the **Bohigas-Giannoni-Schmit (BGS) Conjecture**, which states that the spectral fluctuations of quantised systems whose classical counterparts are chaotic agree with the predictions of Random Matrix Theory (RMT). In this project, you must investigate the distribution of "spacings" between adjacent energy levels $s_i = E_{i+1} - E_i$.

For integrable systems (e.g., a rectangular billiard), the energy levels are uncorrelated. The probability density of spacings follows a Poisson distribution:

$$P(s) = e^{-s}$$

This implies a high probability of finding near-degenerate levels ($s \to 0$).

For chaotic systems with time-reversal symmetry (e.g., a Bunimovich Stadium), the levels "repel" each other. The spacing distribution follows the Wigner-Dyson distribution:

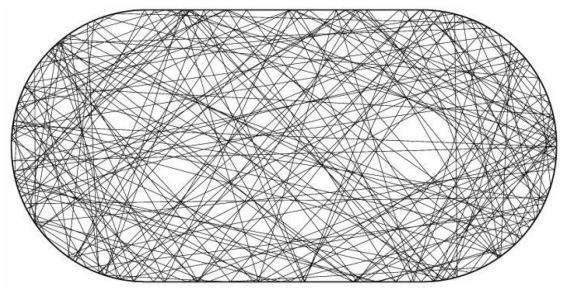
$$P(s) = \frac{\pi}{2} s \exp\left(-\frac{\pi}{4} s^2\right)$$

Note that $P(s) \to 0$ as $s \to 0$, indicating **Level Repulsion**.

2.1 The Billiard Geometry

Students will numerically solve the time-independent Schrödinger equation (the Helmholtz equation) for two geometries:

1. **The Integrable Case:** A rectangle with side lengths L_x and L_y (check both the cases in which L_x/L_y is irrational or rational).
2. **The Chaotic Case:** A Bunimovich Stadium (two semi-circles joined by a rectangle).



Use a finite-difference scheme or the **Boundary Element Method** to find the first 500–1000 eigenvalues (k_n^2).

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)\psi + k^2\psi = 0, \quad \psi|_{\partial\Omega} = 0$$

2.2 Deliverables

- **Spacing Histograms:** Plot $P(s)$ for both geometries and fit them to the Poisson and Wigner-Dyson curves.
- **Visualisation:** Plot the spatial probability density $|\psi(x, y)|^2$ for high-energy states. Identify any "scars" of classical orbits in the chaotic billiard.

3 Pattern Formation in Reaction-Diffusion Systems

In 1952, Alan Turing proposed that a system of reacting and diffusing chemicals could spontaneously break symmetry to form stable spatial patterns. This counter-intuitive phenomenon, where diffusion (usually a homogenizing force) destabilizes a uniform state, provides a fundamental model for biological morphogenesis. In this project, you will analytically derive the conditions for Turing instability and numerically simulate the Gray-Scott model to produce a variety of complex patterns. Consider two chemical species, an activator u and an inhibitor v. Their concentrations $u(\mathbf{x}, t)$ and $v(\mathbf{x}, t)$ are governed by the following system of coupled partial differential equations (PDEs):

$$\frac{\partial u}{\partial t} = D_u \nabla^2 u + f(u, v)$$
$$\frac{\partial v}{\partial t} = D_v \nabla^2 v + g(u, v)$$

where D_u and D_v are the diffusion coefficients and the functions f, g describe the local reaction kinetics.

3.1 Theoretical Analysis

To find the conditions for pattern formation, follow these steps:

1. **Steady State:** Identify the homogeneous fixed point (u^*, v^*) where $f(u^*, v^*) = g(u^*, v^*) = 0$.
2. **Jacobian Matrix:** Linearize the system around (u^*, v^*). Let $\mathbf{w} = (u - u^*, v - v^*)^T$. In the absence of diffusion, the stability is determined by the Jacobian

$$J = \begin{pmatrix} f_u & f_v \\ g_u & g_v \end{pmatrix}$$

   where subscripts denote partial derivatives evaluated at the steady state.
3. **Turing Conditions:** Prove that for a Turing instability to occur, the following must hold:
   - $f_u + g_v < 0$ (Stable without diffusion)
   - $f_u g_v - f_v g_u > 0$ (Stable without diffusion)
   - $D_v f_u + D_u g_v > 0$ (Diffusion-driven instability)
   - $(D_v f_u + D_u g_v)^2 - 4D_u D_v (f_u g_v - f_v g_u) > 0$
4. **Diffusion Ratio:** Show that these conditions imply $D_v > D_u$.

3.2 Computational Analysis

For the computational component, use the Gray-Scott kinetics, which represent a "feed-kill" system:

$$f(u, v) = -uv^2 + F(1 - u) \tag{1}$$
$$g(u, v) = uv^2 - (F + k)v \tag{2}$$

where F is the feed rate of u and k is the kill rate of v. Implement a 2D numerical solver using a Finite Difference scheme on a grid of size $L \times L$ (e.g., 128×128) with periodic boundary conditions. To do so, use the 5-point stencil for the ∇^2 operator with grid spacing h:

$$\nabla^2 u_{i,j} \approx \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j}}{h^2},$$

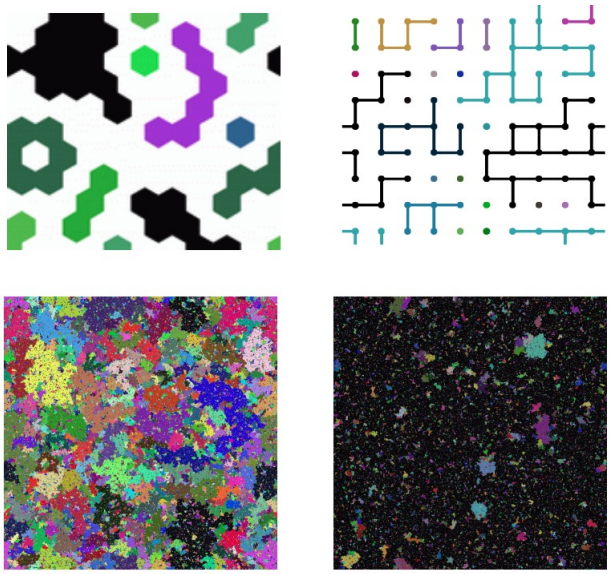
as well as the Forward Euler method for time stepping. Ensure the time step Δt satisfies the stability criterion

$$\Delta t < \frac{h^2}{4 \max(D_u, D_v)}$$

1. **Simulation:** Initialize the system with $u = 1, v = 0$ and add random noise to a small central region.
2. **Parameter Exploration:** Using $D_u = 0.16$ and $D_v = 0.08$, explore the (F, k) space. Identify parameters that yield
   - **Spots**
   - **Stripes/Labyrinths**
   - **Solitons**
3. **Analysis:** Plot the final concentration of v as a heatmap and discuss how the diffusion ratio influences the characteristic wavelength of the pattern.

4 Percolation, fractals, and fires.

Percolation was first introduced by Broadbent in 1954 by asking if I have a lattice, how many connections can you cut before removing the possibility of connecting two distant points of the network.



Let's think of a forest as a square grid in which each cell is occupied by a tree with probability p, or empty with probability $1-p$. This distribution of trees and gaps will be the initial state. Imagine that one of the trees is on fire and it transfers to all next neighbours. The density of trees will determine if the fires propagates through the forest or it does not. If we want to maximise the trees density without having fire percolation we may determine the critical probability p_c that makes the system percolate.

In this project, you should generate square and triangular lattices of size up to $L = 512$ or higher in order to study the percolative and fractal properties of the aggregates [1].

- For different system sizes calculate $p_c(L)$. Calculate it in the thermodynamical limit $L \to \infty$.
- Estimate the probability distribution function of cluster sizes in the regime $p < p_c$, $p = p_c$, and $p > p_c$.
- For $p = p_c$ calculate the fractal dimension and study the scale invariance of the system.

Once you have analised static lattices you must study dynamical ones. To do so use the same topologies as in the previous case, but define the probability p as the probability of growing a tree per unit of time. Then, we define also the probability of a lightning that hits a random tree as p_l per unit of time. This will create dynamical fires. Study the following.

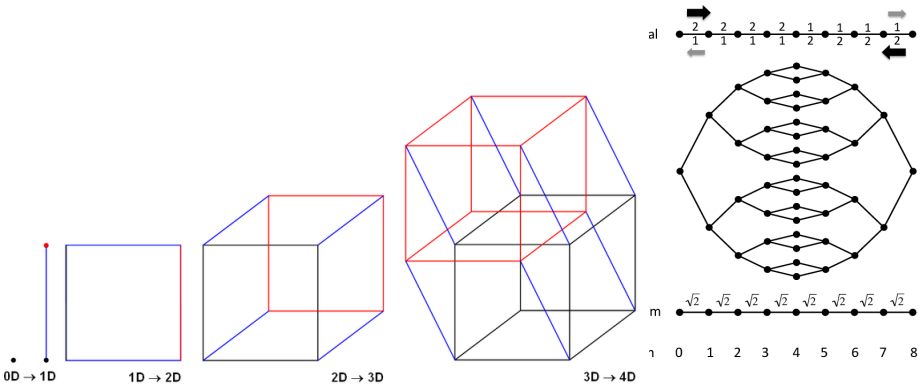
- Calculate the size of the fires (as number of trees) and their probability, as a function of both p and p_l. Plot it in a log-log scale.
- Is there any phase transition here? If yes, characterise it as a function of the ratio p/p_l.

[1] `https://ergodic.ugr.es/cphys/LECCIONES/percolacion/percolacion.pdf`

5 Random walks in complex graphs.

A random walk in a graph is just an agent that can jump from node i to node j if they are connected. In the general case, different probabilities may be given to each node. In this project you must study two examples of random walks in complex graphs.

A hypercube (HC), also known as an n-dimensional cube or an n-cube, is a graph that extends the concept of a square and a cube to higher dimensions. The term "hypercube" is typically used when referring to four or more dimensions, although in a broader sense, a line segment (1-cube), square (2-cube), and cube (3-cube) are also considered hypercubes in their respective dimensions. Another example of complex networks are "glued binary trees" (GBT). A binary tree is a graph that starts with a root and each node has two descending one. A glued binary tree of depth N is a tree that increases $N/2$ layers and then decreases until forming a new root.



In this project we are going to study the performance of a random walker that starts in one node of a graph (HC or GBT) and wants to travel to the opposite one. To define the performance we are going to use the concept of "hitting time", that is the expected time of travelling from node i to node j [1].

$$\tau_{ij} = \sum_{t=0}^{\infty} t\, p_j(t).$$

In this project you should:

- Calculate, by combinatorial analysis, the hitting time between opposite vertices in HC and GBT as a function of their dimension/depth.
- Make a numerical simulation to analyse the hitting time between arbitrary nodes and study it as a function of the dimension/depth and the nodes separation.

[1] H. Krovi and T.A. Brun. Hitting time for quantum walks on the hypercube. Phys. Rev. A **73**, 032341 (2006).

6 Quantum walks in a Lattice.

A "discrete quantum walk" represents the quantum analog of the classical random walk, incorporating the principles of quantum mechanics called superposition. In classical random walks, the walker moves randomly at discrete time steps to adjacent positions on a predefined structure, like a line or lattice, with the direction determined by some probabilistic rule. In contrast, a discrete quantum walk operates on a quantum system, where the walker's position is described by a quantum state [1,2].

Mathematically, we define the state of the walker in a lattice as a vector in a Hilbert space $\mathcal{H}$. This space is divided in the "coin" space, that decides the direction of the movement, and the "position" space. $\mathcal{H} = \mathcal{H}_C \otimes \mathcal{H}_P$. $\mathcal{H}_C$ is spanned by four basis states $\{|\uparrow\rangle, |\downarrow\rangle, |\leftarrow\rangle, |\rightarrow\rangle\}$, while the position Hilbert spaces is spanned by the position $\{|i,j\rangle\}$. The evolution of the system is describe by the following unitary operator:

$$
\begin{aligned}
S = &|\uparrow\rangle\langle\uparrow| \otimes \sum_j |i, i+1\rangle\langle i, j| + |\downarrow\rangle\langle\downarrow| \otimes \sum_j |i, j-1\rangle\langle i, j| \\
&+ |\leftarrow\rangle\langle\leftarrow| \otimes \sum_i |i+1, j\rangle\langle i, j| + |\rightarrow\rangle\langle\rightarrow| \otimes \sum_i |i-1, j\rangle\langle i, j|
\end{aligned}
$$

As the state of the coin "changes" (think about this!) after each steps, it may be rotated in order to ensure the unbiasedness of the walker. To do so we apply first an operator in the coin space. Many operators can do it as, for instance, the Hadamard coin

$$
H = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.
$$

The general evolution operator is then given by

$$
U = S(C \otimes \mathbb{I}).
$$

In this project we will study the performance of a quantum walk in a lattice by doing the following:

- Study analytically the evolution of a quantum walk for a small number of steps.
- Calculate the probability distribution of position at different times $t = 10,\ 100,\ 1000$. Compare with the classical case.
- Make the same analysis for a different coin operator.

[1] Quantum random walks: an introductory overview. Contemporary Physics **44**, 307 (2003)
[2] Controlling discrete quantum walks: coins and initial states. New J. Phys. **3**, 83 (2003).

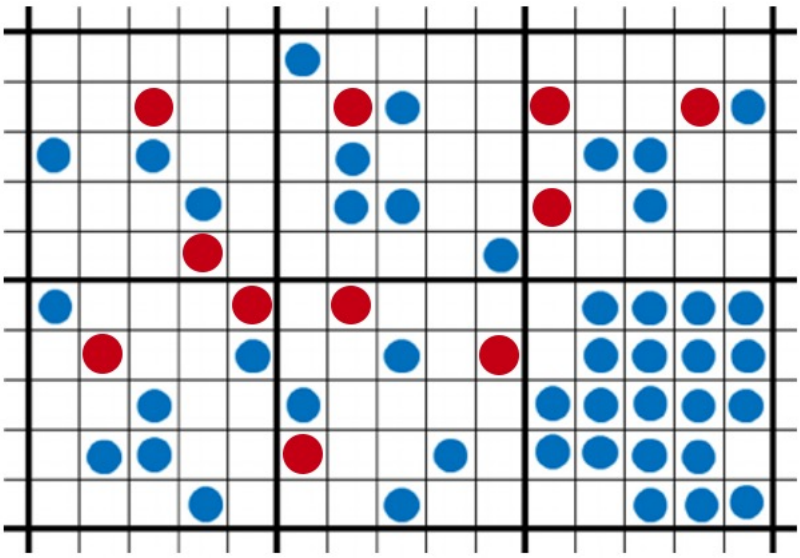
7 The Information-Limited Maxwell Demon

In 1961, Rolf Landauer asserted that "Information is Physical." He argued that any logically irreversible transformation of information, such as the erasure of a bit, must be accompanied by a corresponding increase in the entropy of the environment. This project explores the physical cost of information processing by simulating a Maxwell Demon with finite memory.

7.1 The System

The gas consists of a 2D discrete lattice with two species:

- **Type A (Cold):** Jumps to a nearest neighbor with probability P_{slow}.
- **Type B (Hot):** Jumps to a nearest neighbor with probability $P_{fast} > P_{slow}$.



7.2 Model: The Memory-Constrained Demon

The Demon monitors the central partition but possesses a finite memory buffer of size M bits.

1. **Measurement:** Each time the Demon identifies a particle and decides to open/close the gate, it records the action in its memory, increasing the Demon's Shannon entropy:

$$H = -\sum p(s) \log_2 p(s)$$

2. **Erasure (The Landauer Step):** Once the memory M is full, the Demon must reset its memory to a "zero state." This process dissipates heat $Q \geq k_B T \ln 2$ per bit into the thermal reservoir.

7.3 Deliverables

- **Simulation:** Model the pressure P_L and P_R over time for different memory capacities M, particle probabilities, and densities.
- **Thermodynamic Accounting:** Track the decrease in gas entropy $|\Delta S_{gas}|$ and compare it to the entropy generated by memory erasure $S_{erase} = Q/T$.
- **Discussion:** Investigate the *Information-to-Work* conversion efficiency. Can the Demon produce enough pressure to perform mechanical work exceeding the cost of its own memory erasure?

8 Phase transitions in the Ising model

The Ising model is a mathematical model used in statistical mechanics to describe ferromagnetism in statistical physics. Developed by the physicist Ernst Ising and his advisor Wilhelm Lenz in the early 1920s, the model consists of discrete variables that represent magnetic dipole moments of atomic spins, which can be in one of two states (+1 or -1). These spins are arranged on a lattice, and the model considers the interaction between nearest neighbours, subject to an external magnetic field. The simplicity of the Ising model lies in its binary state system, yet it captures the essence of magnetic phase transitions, such as the transition from a magnetised (ordered) state to a non-magnetised (disordered) state at a certain temperature known as the critical temperature.

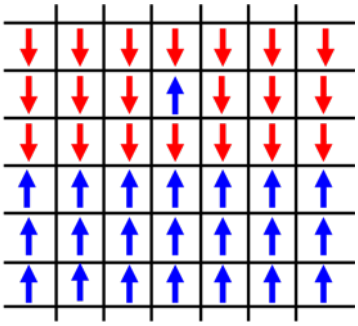
The Ising model is defined by an interaction spin Hamiltonian

$$H = -J_1 \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i,$$

where $J_!$ is the coupling strength, h a transversal magnetic field, and the expression $\langle i, j \rangle$ represents nearest neighbours indexes. If we include a long-range distance, we can take into account next-nearest neighbours interactions.

$$H = -J_1 \sum_{\langle i,j \rangle} s_i s_j - J_2 \sum_{\langle\langle i,j \rangle\rangle} s_i s_k - h \sum_i s_i,$$

where J_2 is the new long-range coupling and $\langle\langle i, j \rangle\rangle$ states for next-nearest indexes.



In this project, we will study the Ising model and its phase transitions. To do so, you have to simulate the random behaviour of such a model for the canonical next-next-neighbours interaction as well for a nearest-neighbours interacting one. To simplify we will take $h = 0$ and $J_1 = 0.5 \cdot J_2 = 1$.

- Calculate the following magnitudes for increasing size of the lattice and extrapolate to the thermodynamical limit. Compare both models for different temperatures
  - Average magnetisation

$$m_N = \left| \frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} s(i,j) \right|.$$

  - Average energy

$$e_N = \frac{\langle E(S) \rangle}{2N}.$$

  - Specific heat capacity

$$c_N = \frac{1}{N^2 T} \left[\langle E(S)^2 \rangle - \langle E(S) \rangle^2 \right].$$

- Compare the results for the numerical simulations with the exact solution from Onsager.
- Estimate the critical temperature by evaluating the specific heat for different sizes and extrapolating to $N \to \infty$.
- Study about the concept of critical points and calculate at least two of them.

[1] https://ergodic.ugr.es/cphys/LECCIONES/ising/ising-SLIDES.pdf

9 Quantum and classical correlations via the CHSH Inequality

Consider an experiment where Alice and Bob share an entangled pair of particles. Alice can choose to measure one of two observables $\{A_1, A_2\}$ and Bob chooses from $\{B_1, B_2\}$. Each measurement yields a result $a_i, b_j \in \{+1, -1\}$. The CHSH correlation value S is defined as:

$$S = \langle A_1 B_1 \rangle - \langle A_1 B_2 \rangle + \langle A_2, B_1 \rangle + \langle A_2, B_2 \rangle$$

where $\langle A_i B_j \rangle$ is the expectation value of the product $a_i b_j$. It can be probed that for any probability distribution $p(a, b)$, the average $|S| \leq 2$.

In quantum mechanics, by using the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ it can be probed that $S = 2\sqrt{2} \approx 2.828$ (Tsirelson's Bound). This is performed by measuring in a basis determined by an angle θ s.t.

$$|a_0\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle$$
$$|a_1\rangle = -\sin\theta|0\rangle + \cos\theta|1\rangle$$

for Alice, and

$$|b_0\rangle = \cos\theta|0\rangle - \sin\theta|1\rangle$$
$$|b_1\rangle = \sin\theta|0\rangle + \cos\theta|1\rangle$$

for Bob.

In this project you have to do the following tasks

- By using the following optimal settings:
  - Alice: $\theta_{A_1} = 0$, $\theta_{A_2} = \pi/4$
  - Bob: $\theta_{B_1} = \pi/8$, $\theta_{B_2} = -\pi/8$

  Show that these settings yield $S = 2\sqrt{2} \approx 2.828$, strictly violating the CHSH inequality.
- Construct a Monte Carlo simulation of a "Classical Universe":
  1. Define functions $a(\theta)$ and $b(\theta)$ that return ± 1.
  2. Run 10^5 random trials to calculate S. Attempt to find *any* local function that exceeds $S = 2$.
- Simulate the quantum experiment using a state-vector approach:
  1. Prepare the density matrix $\rho = |\Phi^+\rangle\langle\Phi^+|$.
  2. Define measurement operators $P(\theta) = |\psi(\theta)\rangle\langle\psi(\theta)|$ where $|\psi(\theta)\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle$.
  3. Calculate the probabilities $P(a, b|\theta_a, \theta_b)$ and sample outcomes to estimate the correlations.
  4. Plot S as a function of the relative angle ϕ between Alice and Bob's detectors.